ECE 308 Lab 3: Servo Position Control

Javier Aldana, Garrison Thompson, Victor Merz
aldana0@purdue.edu | thomp901@purdue.edu | merz4@purdue.edu
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1. Introduction

Servo position control is an industry standard control system that ensures positional accuracy for motors. It consists of a negative feedback loop system that uses motor encoders to read positional slop and sends out signals to the encoder to move towards its intended position.

We first used step response tests to show how properly designed controllers impact a system to meet desired specifications. This part of the experiment tests a designed controller to ensure desired behavior and test any environmental issues, such as saturation issues. This was done by first testing our desired controller in a Simulink simulation and then implementing it into our SRV02 hardware.

The next section of the lab was about using ramp response tests to measure steady state error and then using those results to compare PV and PIV controllers. As with the first section of the lab, the first step was to use a simulation to confirm desired behavior and then using the hardware to compare and contrast how each controller tracks a ramp input. Ideally, this is to show how a PIV controller can result in zero steady state error where a PV controller cannot.

2. Results

	PV Simulation	PV w HP Filter Simulation	PV Hardware Experiment	PV Hardware Experiment (0.5%)
Steady State Value	0.3927	0.3927	0.3927	0.3927
Peak Value	0.4111	0.4136	0.4280	0.3958
Peak Time	209.244ms	209.244ms	120.088ms	127.329ms
Percent Overshoot	4.69%	5.32%	8.99%	0.78%

Table 3.1: Measured parameters from step response

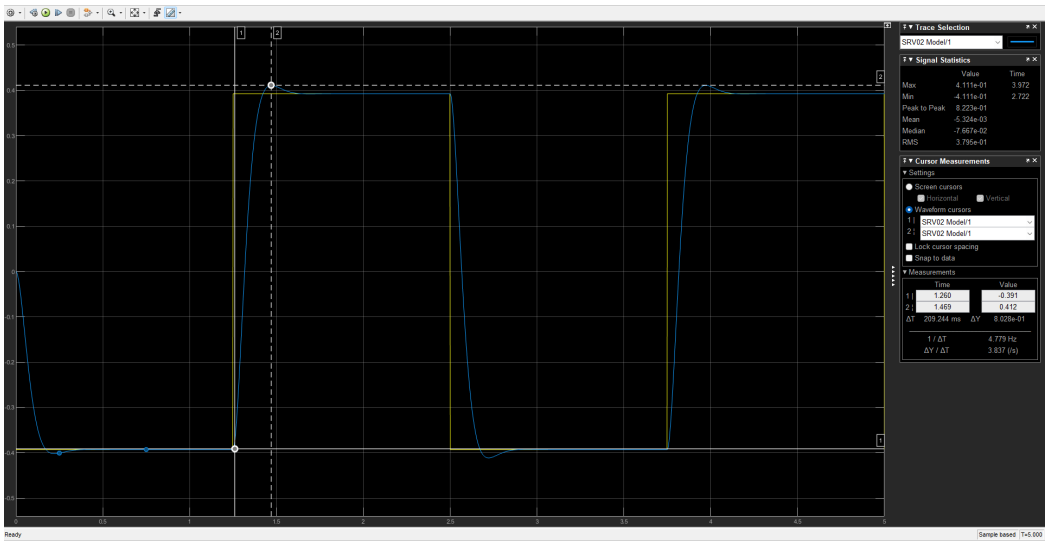


Figure 1: Simulink Step Response using PV controller (5% overshoot). X axis: time (sec), Y axis: motor position (rad), yellow trace: reference signal, blue trace: actual output response

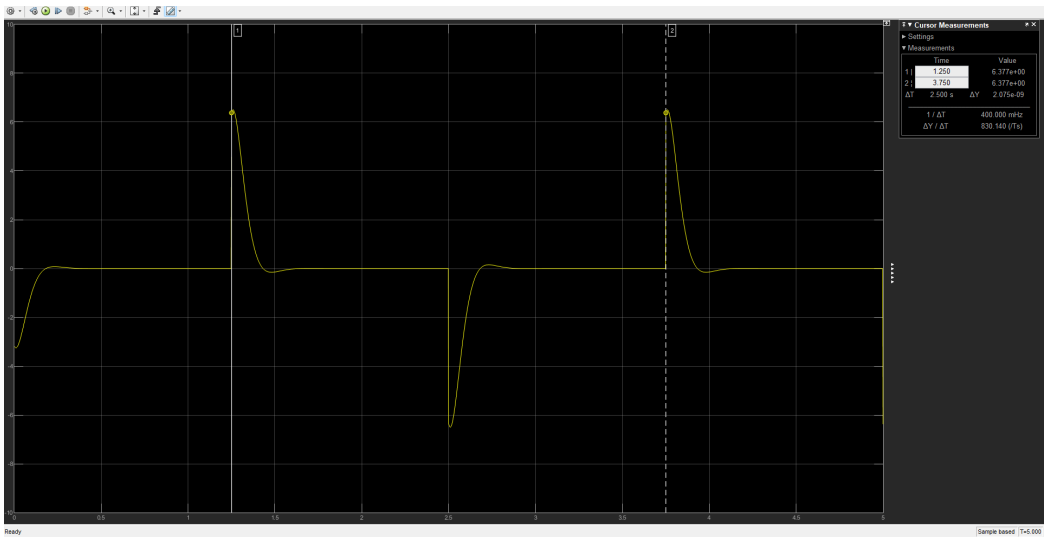


Figure 2: Input Voltage to PV controller (5% Overshoot). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

The response time is the difference in x axis from the rise of the yellow input reference signal and the highest peak on the step response, the input voltage is the necessary actuation signal to reach that step response. The percent error for this simulation was below 5%, but the rise time was greater than 200ms, so it did not meet spec in that regard.

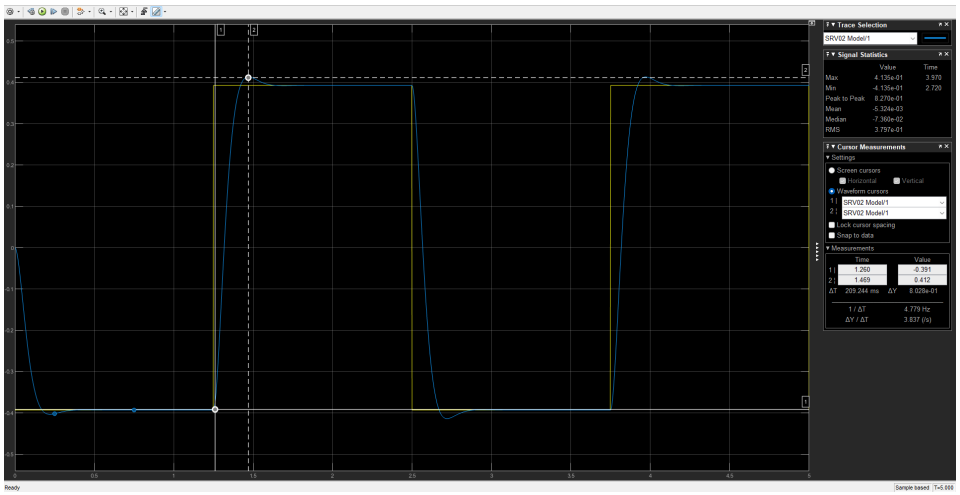


Figure 3: Simulink Step Response using PV controller (High Pass Filter). X axis: time (sec), Y axis: motor position (rad), yellow trace: reference signal, blue trace: actual output response

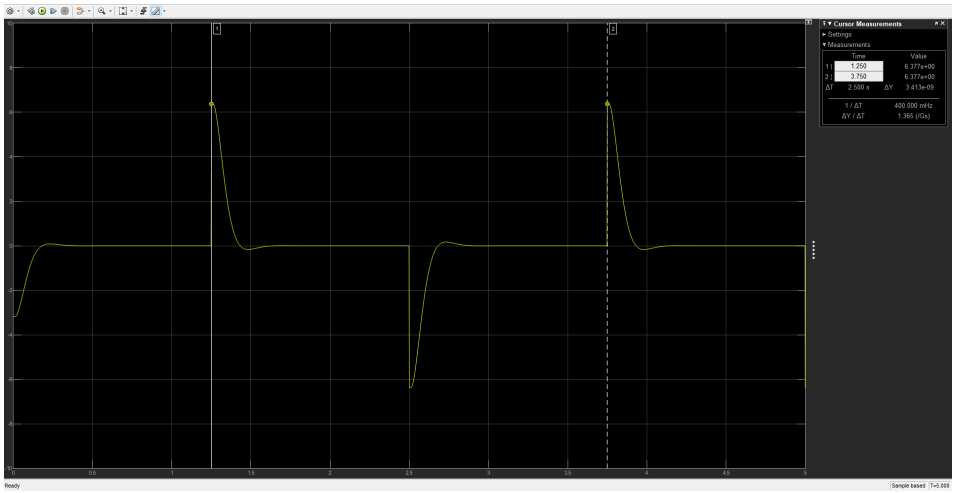


Figure 4: Input Voltage to PV controller (High Pass Filter). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

The percent error for this simulation was above 5%, the rise time was greater than 200ms, so this simulation did not meet the spec in either regard. The effect of the high pass filter increased the error, however a real hardware application could have high frequency noise that would make the first controller worse but not to the high pass filter.

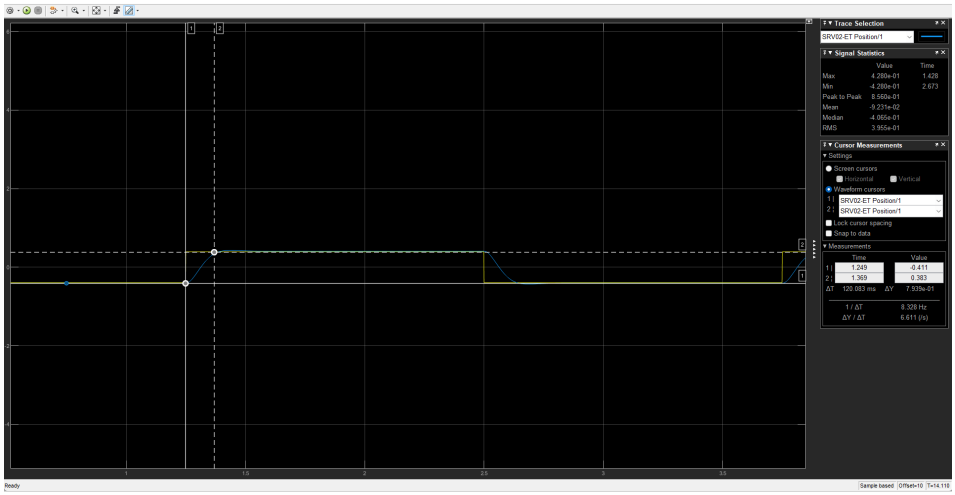


Figure 5: Hardware Step Response using PV controller. X axis: time (sec), Y axis: motor position (rad), yellow trace: reference signal, blue trace: actual output response

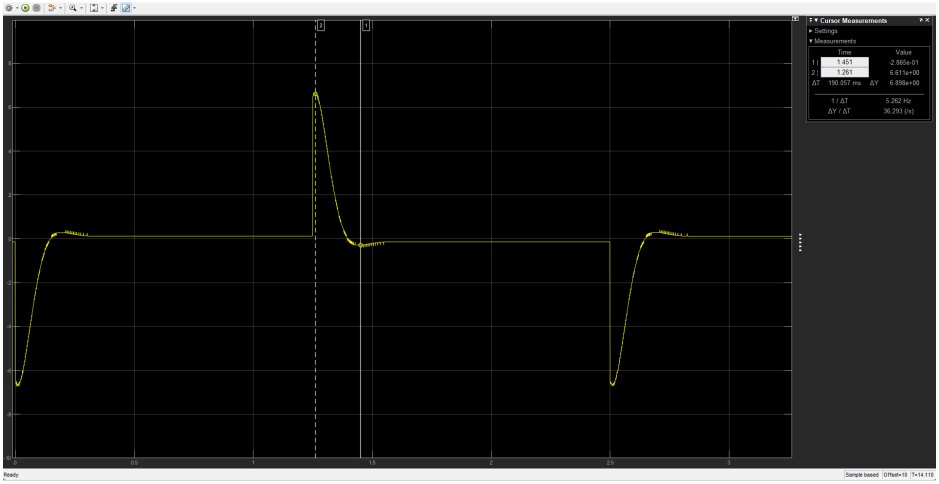


Figure 6: Input Voltage to PV controller (Hardware Experiment). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

The percent error for this simulation was around 9% which is significantly above 5%, however the rise time was below 200ms. Therefore, the spec was met on one of the two requirements. We saw a significantly faster rise time in the hardware experiment as opposed to the simulation.

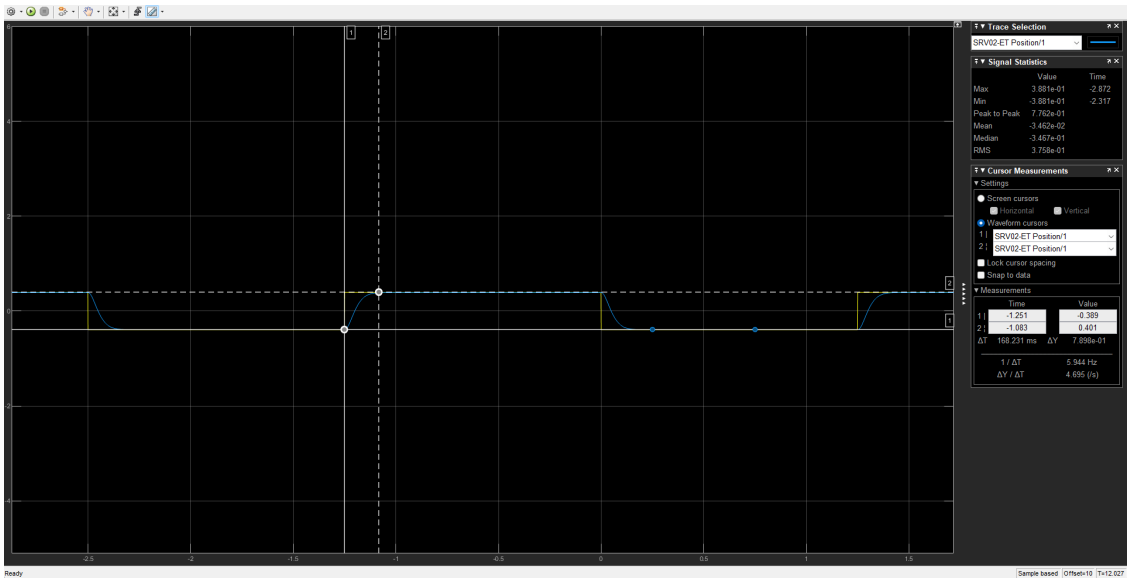


Figure 7: Hardware Step Response using PV controller (0.5% Overshoot). X axis: time (sec), Y axis: motor position (rad), yellow trace: reference signal, blue trace: actual output response

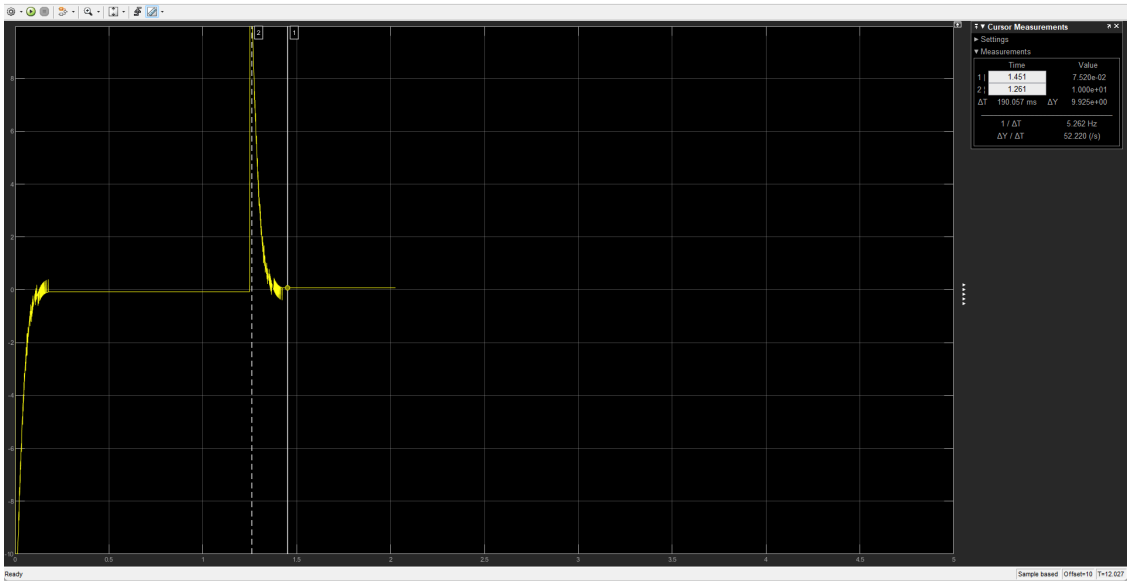


Figure 8: Input Voltage to PV controller (Hardware Experiment 0.5% Overshoot). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

The percent error for this step response was -1.14%, which is significantly bigger in magnitude than 0.5%. This is most likely due to slightly incorrect kvalues, we used ka = 16.34, kv = 0.306. When discussing with the TA, she gave us new values, and the new graphs are below.

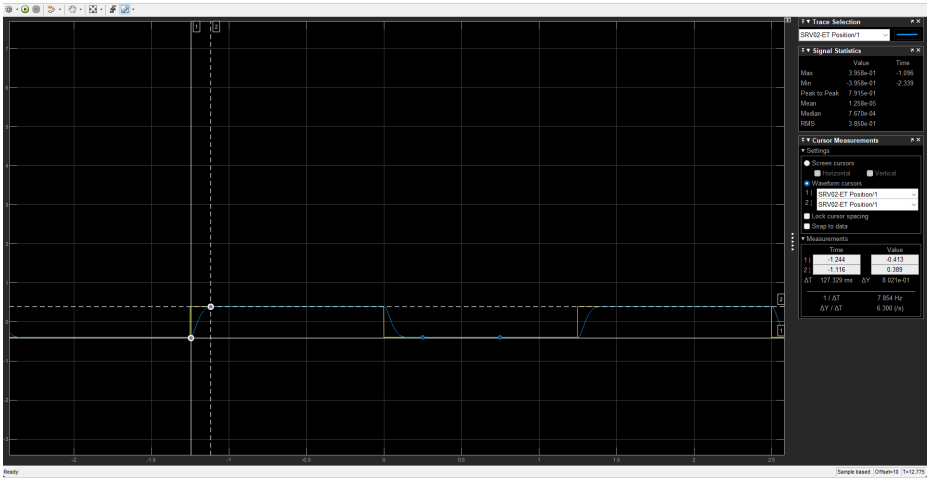


Figure 9: Hardware Step Response using PV controller (0.5% overshoot with TA Values). X axis: time (sec), Y axis: motor position (rad), yellow trace: reference signal, blue trace: actual output response

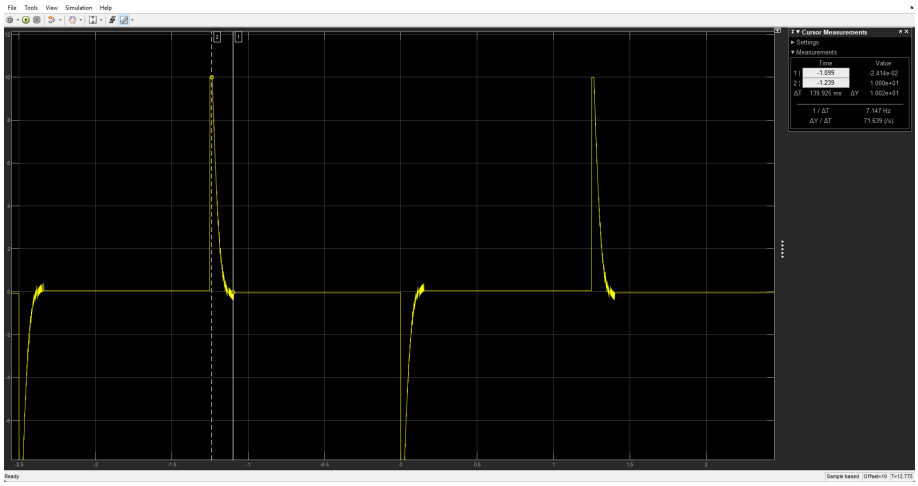


Figure 10: Input Voltage to PV controller (Hardware Experiment 0.5% overshoot with TA Values). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

The percent error for this new step response is 0.78%, significantly smaller than the last percent error of -1.1%, also noticeably it is a percent error in the positive direction, not negative. We still have not met the spec of within 0.5% however we have not determined what has led to that. The values in Chart 3.1 come from this graph.

Values: ka = 15.74, kv = 0.226

	PV Simulation	PV Hardware Experiment	PIV Hardware Experiment
Steady-state Error	-9.48%	-7.84%	+11.2%

Table 3.2: Steady State Error for a Ramp Reference

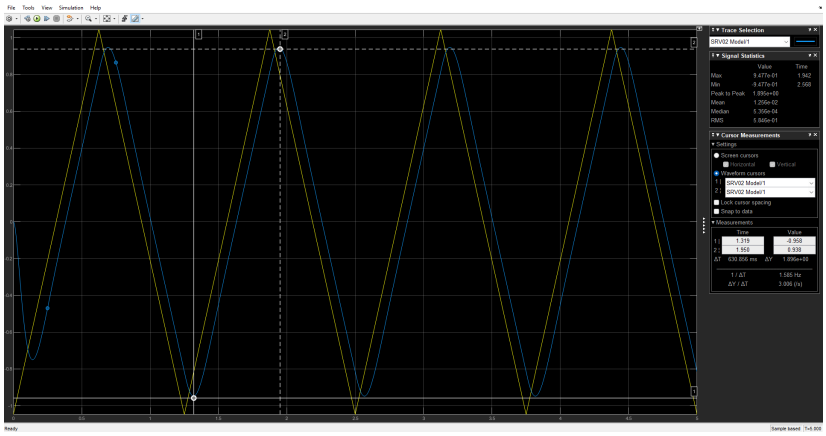


Figure 11: Simulink Ramp Response using PV controller. X axis: time (sec), Y axis: motor position (rad), yellow trace: reference signal, blue trace: actual output response

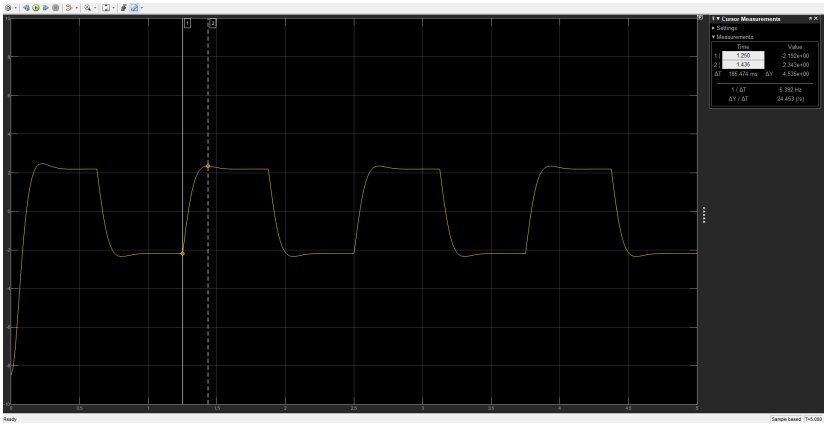


Figure 12: Input Voltage to PV controller Ramp Response. X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

Max = .9477, input = 1.047, error = -9.48%

There is an observable negative error in the ramp response as well as a lag in phase

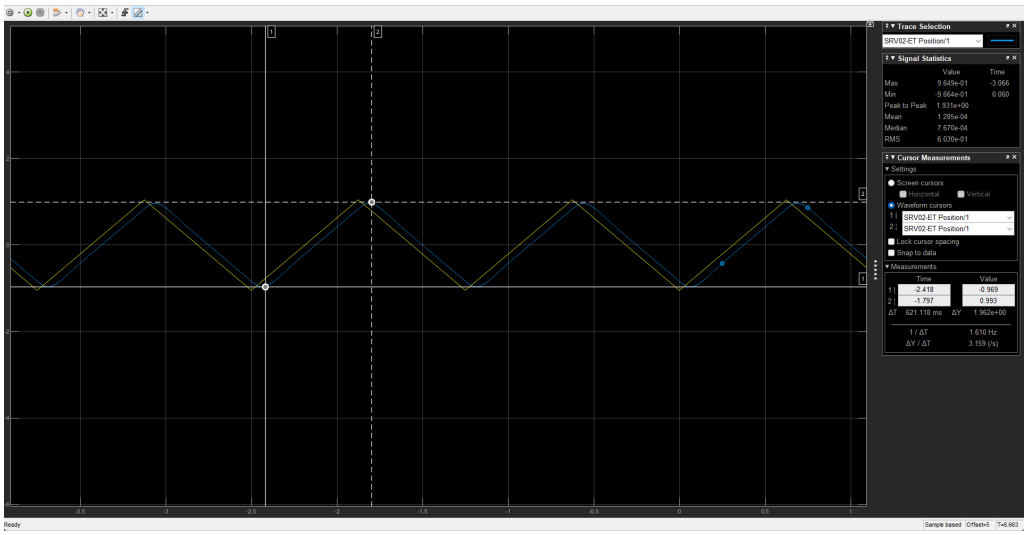


Figure 13: Hardware Ramp Response using PV controller. X axis: time (sec), Y axis: motor position (rad), yellow trace: reference signal, blue trace: actual output response

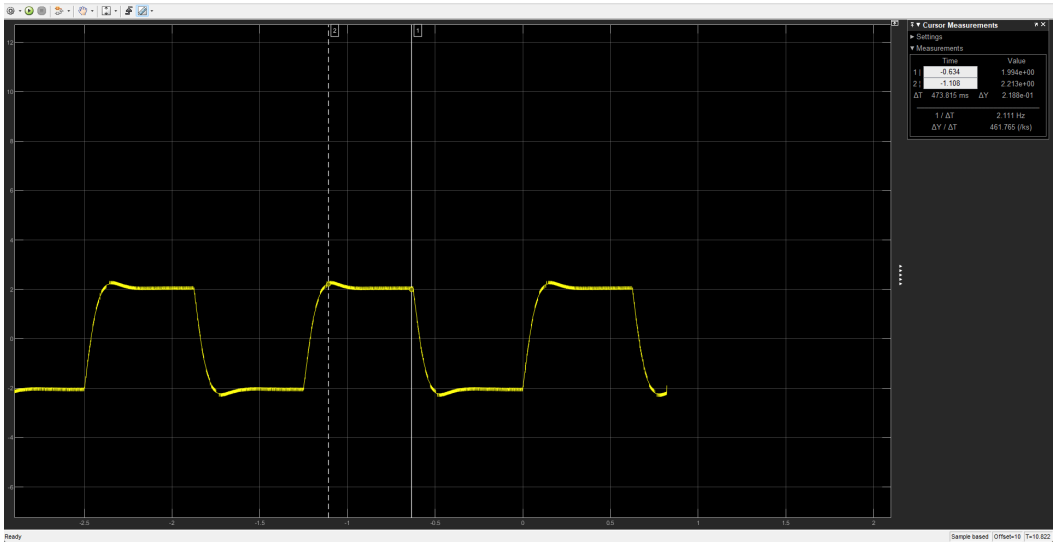


Figure 14: Input Voltage to PV controller Ramp Response (Hardware). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

Max = .9649, input = 1.047, error = -7.84%

The step response signal is somewhat closer to the input triangle wave, but there is still a negative error. The lag to the input wave is like the simulated response.

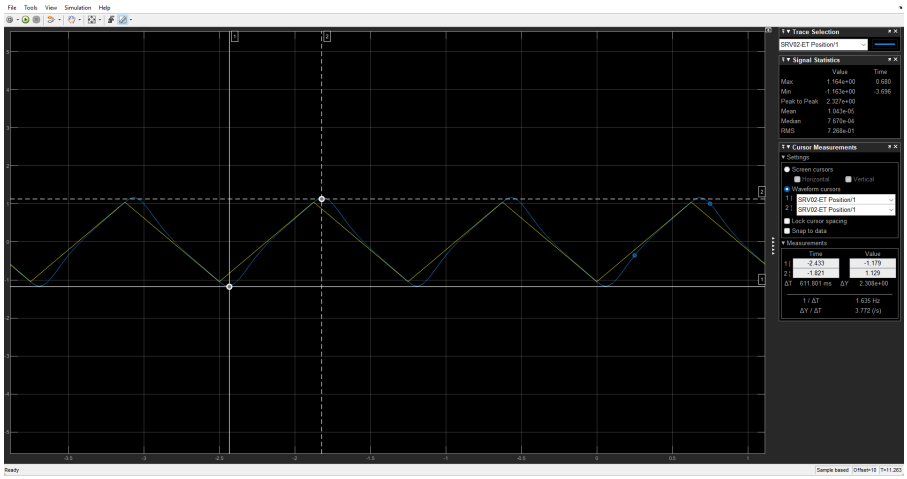


Figure 15: Hardware Ramp Response using PIV controller. X axis: time (sec), Y axis: motor position (rad), yellow trace: reference signal, blue trace: actual output response

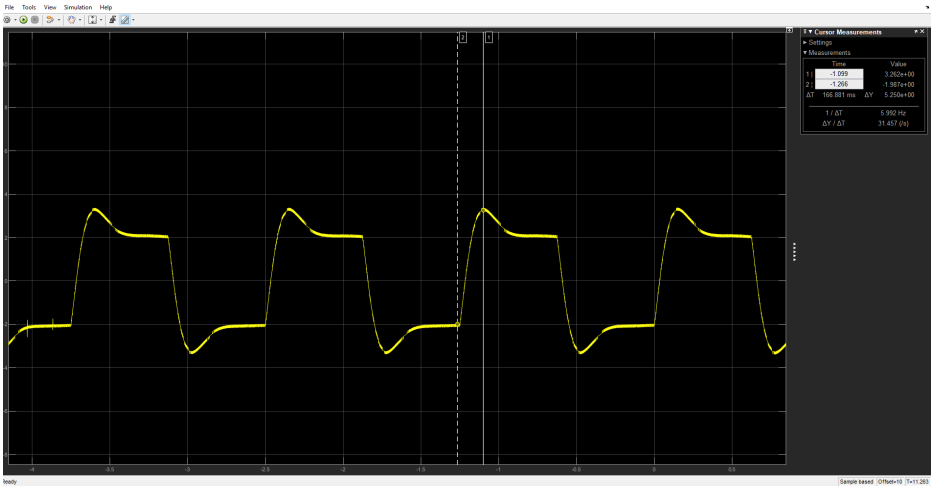


Figure 16: Input Voltage to PV controller Ramp Response (Hardware). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

Max = 1.164, input = 1.047, error = +11.2%

The PIV controller made the error go from negative to positive, there is a similar lag as in the PV controller.

3. Conclusions

The PV and PIV controllers both worked to an extent. However, as shown in Table 3.2, the PV controller had a relatively low steady state error, which was negative. This implies that the actual load position is greater than the desired position, which means this resulted in overshooting. For PIV, the steady state error was positive, meaning the actual load position fell short of the target.

While, in theory, the PIV controller is supposed to result in a lower steady state error, there are a few things that could explain the slightly higher error, such as the high-pass filter having adverse effects. Regardless, the fact that the sign of the steady state error shifted implies that the gain ki did impact the controller, which does align with our theory.